

MATRIC NUMBER									

DC014
Sains Komputer 1
Semester I
Sesi 2023/ 2024
2 jam

NAME: _____

PRACTICUM: _____

KOLEJ MATRIKULASI JOHOR
KEMENTERIAN PENDIDIKAN MALAYSIA
JOHOR MATRICULATION COLEGE
MISTRY OF EDUCATION MALAYSIA

PRA PSPM 1
PROGRAM MATRIKULASI SISTEM EMPAT SEMESTER
PRE -PSPM 1
MATRICULATION PROGRAMME FOUR SEMESTER SYSTEM

SAINS KOMPUTER 1
2 JAM

JANGAN BUKA SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

ARAHAN KEPADA CALON:

Kertas ini mengandungi 6 soalan.
Jawab semua soalan di ruang yang disediakan.

INSTRUCTION TO CANDIDATE:

This booklet consists of 6 questions.
Answer all questions in the space provided.

UNTUK KEGUNAAN PEPERIKSA		
No. Soalan	Pemeriksa	KKP/KP
1		
2		
3		
4		
5		
6		
Jumlah		

Kertas soalan ini mengandungi 11 halaman bercetak, termasuk muka hadapan.
This booklet consists 11 printed pages, include cover page.

1. Multimedia is an interactive media and provides multiple ways to represent information to the user in a powerful manner.
- a) State and explain **1 (one)** element that has a powerful impact in multimedia to the audience.

[2 marks]

- b) Identify the multimedia element based on the situation above:

[2 marks]

Situation	Multimedia element
Malek inserted speech and music into his multimedia project to make the project become real.	
Salman designed a 3D interactive project for primary school to make a presentation more attractive.	

- c) Every object occupies some space, but has a variety of dimensions. Dimension is an attribute for an object which defines the position of the target object. **Figure 1** is a dimension of an object.

[2 marks]

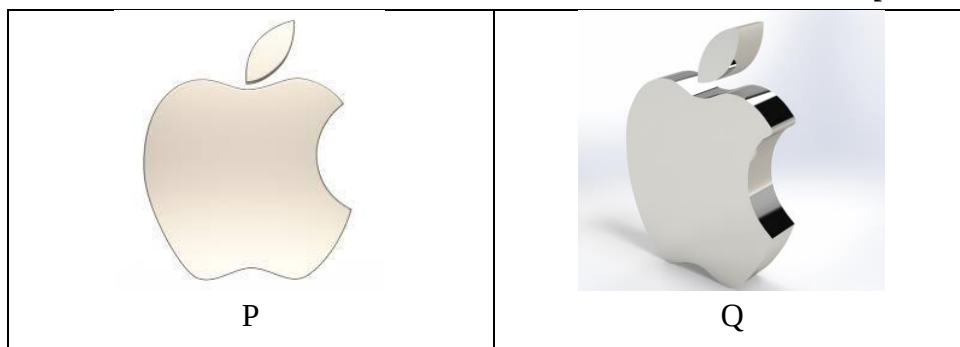


Figure 1

Based on **Figure 1** differentiate P and Q in multimedia application representation in the term of axis

Picture Y: _____

Picture Z: _____

2. a) In order to design multimedia involves three phases, which is every phase has its own activities.

[6 marks]

Identify the phase based on the activities:

Situation	Phase
i. Each object on the screen serves a purpose and communicates a visual message about the content to the user. The goal is to keep users oriented and draw them into the product.	
ii. The final step in the process is distributing your multimedia project.	
iii. Quality control goes on throughout the process. The storyboards are helpful for checking the sequencing.	
iv. The prototype screens and provide the foundation for production.	
v. Illustrations, graphics, buttons, and icons are created using the prototype screens as a guide.	
vi. The original files, including audio, video, and the native software formats, are archived for future upgrades or revisions.	

- b) Multimedia authoring tools give an integrated environment for joining together the different elements of a multimedia production.

Based on the description below, identify Multimedia authoring tools.

[3 marks]

Description	Multimedia authoring tools.
Multimedia elements or event are present and organized with the control of speed at which it plays.	
These tools are best used when the bulk of your content consists of elements that can be viewed individually, letting the authoring systems link.	
The multimedia elements are organized into a structural framework, displayed as flow diagrams with branching paths.	

- c) How does audio navigation benefit individuals with visual impairments, and what are some common examples of tools or technologies that implement this accessibility feature **[3 marks]**

- d) Halim likes to produce a video and post in an Online Social Media, his content is to educate people while watching his post.
Differentiate between online multimedia delivery and offline multimedia delivery **[2 marks]**

Offline	Online

- e) Explain why online multimedia delivery nowadays is more popular compared with offline multimedia delivery. **[2 marks]**

3. Databases are designed to offer an organized mechanism for storing, managing and retrieving information.

- a) Describe **two (2)** advantages of using databases. **[4 marks]**

- b) Identify the component of Database Management System (DBMS), based on the criteria given:

- i) Refers to a collection of programs used by a computer in a database system. This includes the DBMS, the operating system of a computer and network, applications and utility programs.

[1 mark]

- ii) Encompasses all users of a database system. This includes the business analysts, database designers, data administrators (DAs), DBAs, application programmers, and end-users.

[1 mark]

4. Database objects are used to carry out many activities. Based on the statement given, state the appropriate database object to be used.

[4 marks]

Statement	Database Object
Allow user uses form to retrieve and maintain data in a database.	
Allows user to display, and generate in hardcopy form.	
Specifies which records should be retrieved by specifying criteria.	
To arrange data in rows and columns and describe the characteristics of data.	

5. a) A computer network is a group of interconnected computers that communicate with each other to share resources and information. The primary uses of computer networks are to facilitate communication, enable resource sharing, provide remote access, and collaborate.

[3 marks]

Situation	The Use of Network
Dr. Salmi wants to be monitoring in real time the conditions of his patients in critical illness or accident.	
Fatimah watches television program content, where a central operation provides programming to many television stations providers.	
Razi and her family travel from Johor Bahru to Mersing for vacation. He needs to monitor objects or personal movement and get from one location to another.	

b) **Figure 2** is a diagram of network in a Computer Lab

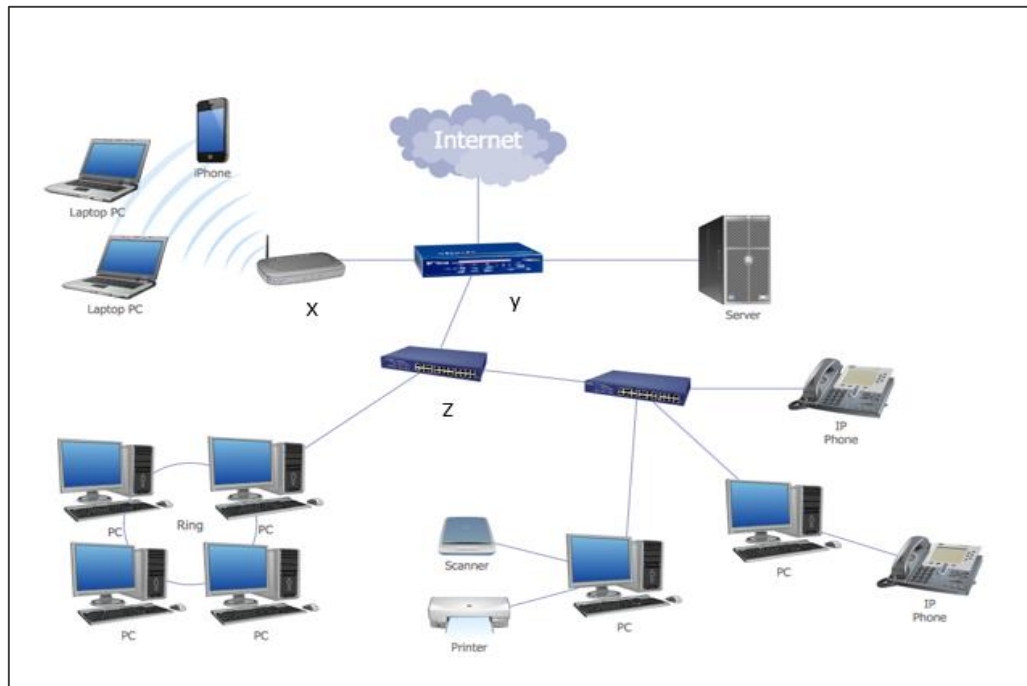


Figure 2

i. Identify and explain the type of transmission media involved in **Figure 2**.

[4 marks]

ii. Identify and explain type of communication device Y and X

[4 marks]

Y	
X	

- c) Identify the wired transmission media based on the situation given:

[3 marks]

Situation	Wired Transmission Media
Shahrul designed a Local Area Network for their school lab, using the cable to reduce noise	
Aminah needs to connect to her satellite television to watch his favorite movie using the ASTRO channel	
Dania wants to connect his home internet for a new Unifi subscriber to a better and faster internet connection	

- d) Identify the wireless transmission media based on the description given:

[3 marks]

Description	Wireless Transmission Media
Navigation system that consists of one or more earth-based receivers that accept and analyse signals sent by satellites in order to determine the receiver's geographic location.	
The part of the invisible spectrum that is contiguous to the red end of the visible spectrum and that comprises electromagnetic radiation of wavelengths	
A mobile phone uses high-frequency radio waves to transmit voice and digital data messages.	

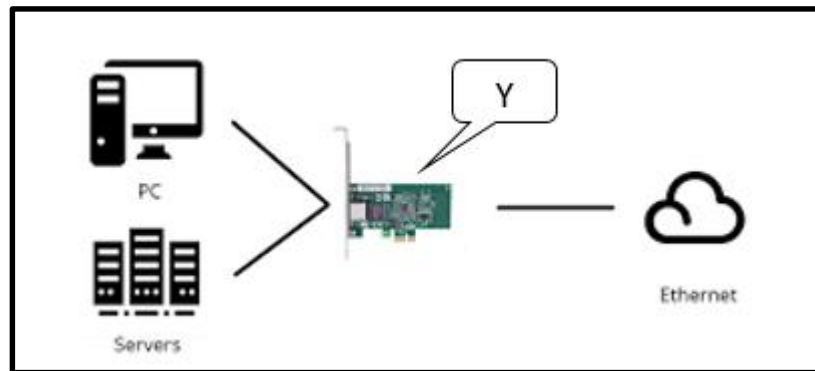


Figure 3

- e) Hamid needs to connect their PC to the internet using network device Y, device Y provides a computer with a dedicated, full-time connection to a network. Identify the component Y, based on **Figure 3**

[1 mark]

6. a) A search strategy can be used to help narrow down a list of hits to a more manageable number. It is possible to find information more effectively and efficiently by using search strategies.

Identify the type of search strategies based on the description given:

[3 marks]

Description	Type of search strategies
To further specify exactly what you want a search engine to find by using AND, OR and NOT.	
Limits the search to a particular search characteristic, such as the page title, URL, page text, top-level domain, or Web site.	
Most search engines automatically list the hits that include all the keywords first, followed by hits matching most of the keywords, continuing down to hits that fit only one of the keywords.	

- b) Web activities can display content, catalogue entries, or categories. When you create a web activity, the action element that you choose determines the type of information to display.

Identify the web activities and the example based on the scenario given.

[8 Marks]

Scenario	web activities	Example
i) Dayang used the web for listening the songs, watching video and movies		
ii) Hizman works as a supervisor at Sendiri Maju Company, he sends reports about the performance of his staff to the Manager in Kuala Lumpur.		
iii) Noryn likes to share their personal or career content, interests, activities, backgrounds or real-life connections.		
iv) Nor Aminah is an influence, she wants her followers to buy the products on online services or over the Internet		

- c) Most people nowadays use the Internet to find specific Information.



Figure 4

- i) Based on the **Figure 4** identify type of Search tool

[1 mark]

ii) Based on answer 6. c) i) explain your answer.

[2 marks]

d) Email (electronic mail) is the exchange of computer-stored messages from one user to one or more recipients via the Internet. Emails are a fast, inexpensive and accessible way to communicate for business or personal use.

Give **two (2)** suggestions to improve email privacy

[2 marks]

e) The Internet of Things (IoT) describes physical objects with sensors, processing ability, software, and other technologies that connect and exchange data with other devices and systems over the Internet or other communication networks.

State **one (1)** example and explain the use of IoT in our personal life.

[2 marks]

f) Green Computing involves reducing electricity and environmental waste. State **two (2)** ways how we can participate in it.

[2 marks]
